

SCREENSENSE: A MACHINE LEARNING FRAMEWORK FOR ANALYZING DIGITAL BEHAVIOR AND PREDICTING WELLBEING FROM SCREEN USAGE PATTERNS

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Abstract- In today's digital environment, prolonged screen exposure, continuous notifications, social media engagement, and irregular daily routines have become common factors that influence concentration, productivity, and emotional wellbeing. Understanding how these digital behaviors affect individuals has become increasingly important, especially among students and young professionals. This paper presents ScreenSense, a machine learning-based digital wellbeing platform developed to study user behavioral patterns and estimate overall wellbeing through intelligent data analysis. The proposed system collects important behavioral attributes such as screen usage duration, application switching frequency, sleep hours, notification activity, social media usage, focus level, mood condition, and anxiety level to understand individual digital habits. Based on these inputs, multiple machine learning algorithms are applied to identify behavioral trends and predict a personalized digital wellbeing score with improved accuracy. In addition to prediction, the system provides interactive visual analytics, behavioral clustering, personalized recommendations, and downloadable analytical reports that help users better understand and improve their digital lifestyle. Experimental evaluation shows that behavioral data can effectively support wellbeing prediction and reveal meaningful relationships between digital habits, focus, and mental wellness. The developed platform offers a practical and scalable solution for promoting healthier technology usage and building awareness of digital wellbeing in modern society.

Index Terms— Digital Wellbeing, Machine Learning, Behavioral Analysis, Screen Time, Productivity, Mental Wellness, Predictive Analytics.

I.INTRODUCTION

The rapid growth of smartphones, social media platforms, and always-connected digital environments has transformed the way individuals learn, communicate, and manage their daily activities. While digital technology offers significant advantages in education, communication, entertainment, and productivity, excessive screen exposure, frequent notifications, continuous application switching, and prolonged social media engagement have introduced new challenges related to focus, emotional stability, sleep quality, and overall mental wellbeing. These behavioral changes are particularly visible among students and young professionals, whose daily routines increasingly depend on digital devices for academic, professional, and social interactions. Although digital technologies improve accessibility and convenience, uncontrolled usage often leads to reduced concentration, increased stress, anxiety, sleep disruption, and decreased productivity, creating a growing concern regarding digital wellbeing in modern society.

Traditional methods for evaluating productivity and mental wellbeing often rely on self-assessment surveys, manual observation, or generalized behavioral studies, which may not accurately capture real-time digital behavior patterns or individual lifestyle differences. In many cases, users remain unaware of how their daily screen habits, notification exposure, and online engagement influence their cognitive performance and emotional health over time differences. In many cases, users remain unaware of how their daily screen habits, notification exposure, and online engagement influence their cognitive performance

and emotional health over time. Existing digital wellness applications primarily focus on screen-time monitoring or application blocking, offering limited analytical insights and lacking personalized predictive capabilities. These limitations highlight the need for an intelligent, data-driven system capable of understanding digital behavior patterns and providing meaningful recommendations based on individual usage characteristics.

To address these challenges, this paper proposes ScreenSense, an intelligent machine learning-based digital wellbeing prediction platform designed to analyze users' digital habits and estimate their overall wellbeing through behavioral analytics. ScreenSense integrates data collection, feature engineering, predictive modeling, and interactive visualization within a unified web-based platform. The system collects important behavioral attributes such as daily screen time, application switching frequency, sleep duration, notification activity, social media usage, focus level, mood condition, and anxiety level to build a comprehensive understanding of user behavior. By analyzing these factors collectively, the platform identifies patterns that influence productivity, concentration, and emotional wellbeing, enabling users to gain deeper awareness of their digital lifestyle.

At the core of ScreenSense are three major functional components: Behavioral Monitoring, AI-Based Wellbeing Prediction, and Personalized Analytics. The Behavioral Monitoring module continuously captures and organizes digital activity data, providing structured insights into users' daily habits. The AI-Based Prediction module applies machine learning algorithms such as Linear Regression, Decision Tree, Random Forest, Support Vector Regression, and Gradient Boosting to estimate an individual's digital wellbeing score with improved predictive accuracy. The Personalized Analytics module presents interactive dashboards, behavioral clustering, trend analysis, and intelligent recommendations that guide users toward healthier digital practices and improved productivity.

To enhance analytical capabilities, ScreenSense also incorporates advanced data science techniques such as K-Means clustering and Principal Component Analysis (PCA) to identify hidden behavioral groups and visualize high-dimensional relationships among digital habits. Personalized profile management, and downloadable PDF reports allow users to track their progress over time while maintaining data privacy and system security.

Experimental evaluation conducted using behavioral datasets demonstrates that machine learning models can effectively identify meaningful relationships between screen usage patterns, sleep quality, notification frequency, emotional state, and digital wellbeing. The results indicate that users with prolonged screen exposure and frequent notification interruptions often experience lower focus levels and increased anxiety, while balanced sleep patterns and controlled digital engagement contribute positively to overall wellbeing. By combining behavioral intelligence, predictive analytics, and user-centered design, ScreenSense provides a scalable and practical solution for promoting healthier digital habits, improving productivity, and supporting mental wellness in an increasingly connected world.

II. METHODOLOGY ADAPTED

The proposed ScreenSense system is developed using an iterative and user-centered development approach, allowing continuous refinement of features based on analytical results and user feedback. The platform follows a modular architecture to ensure scalability, maintainability, and efficient integration of machine learning, analytics, authentication, and reporting services. The backend logic is implemented using Python, while the interactive web interface is developed using Streamlit to provide a responsive and personalized user experience. Behavioral data collected from users is processed through dedicated modules for data cleaning, feature engineering, model training, prediction, and visualization. The system also integrates secure authentication, session management, and PDF-based reporting to deliver a complete end-to-end digital wellbeing platform. Machine learning algorithms are employed to analyze user behavioral patterns and generate real-time digital wellbeing predictions based on personalized inputs.

A. Tech Stacks Used

The development of ScreenSense adopts a modern machine learning-driven software architecture to ensure performance, usability, and analytical accuracy. The frontend is implemented using Streamlit, HTML, CSS, and JavaScript-based notification activity customization to deliver an interactive and visually engaging dashboard. The backend processing is developed using Python, which manages data preprocessing, feature engineering, model execution, user authentication, and report generation. Pandas and NumPy are used for data manipulation and numerical

computation, while Scikit-learn is used for machine learning model development and evaluation. User information, prediction history, and analytical records are stored using SQLite and CSV-based storage for efficient data management. Interactive visual analytics are generated using Plotly and Matplotlib, allowing users to explore behavioral trends and prediction results in real time.

B. Architecture Diagram



Fig. 1. System Architecture

C. Modules Used in ScreenSense

The ScreenSense platform consists of five major functional modules designed to analyze digital behavior and provide personalized wellbeing insights:

1) User Authentication and Profile ManagementThe authentication module provides:

- Secure user registration and login using encrypted password storage
- Session-based access control and personalized profile management
- User-specific prediction history and behavioral tracking

2) Behavioral Data Collection Module

The data collection module captures:

Daily screen time, application switching frequency, and

- Sleep duration, social media usage patterns, focus score, mood score, and anxiety level .

- Structured behavioral inputs through a multi-step wizard interface

3) Machine Learning Prediction Module

The prediction module performs:

- Data preprocessing, normalization, and feature engineering
- Real-time digital wellbeing score prediction using trained models
- Confidence score generation and personalized recommendation output

4) Analytics and Visualization Module

The analytics module provides:

- Interactive dashboards, behavioral trend analysis, and correlation visualization
- Clustering analysis for identifying user behavior groups

- PCA-based dimensional visualization for pattern discovery

5) Reporting and History Management Module The reporting module supports:

- Automated PDF report generation with behavioral insights

- Prediction history storage and progress monitoring

- Personalized recommendations for long-term digital wellbeing improvement

D. Which algorithm are we going to use for digital wellbeing prediction?

To predict users’ digital wellbeing scores, multiple supervised machine learning algorithms are evaluated, including Linear Regression, Decision Tree, Random Forest, Support Vector Regression, and Gradient Boosting. After comparative analysis, the Gradient Boosting algorithm is selected as the final prediction model due to its ability to capture complex behavioral relationships, reduce prediction errors, and deliver high predictive accuracy across diverse user patterns.

E.Why is Gradient Boosting preferred over other algorithms?

The Gradient Boosting algorithm is preferred because it combines multiple weak learners to build a highly accurate predictive model. Unlike simpler regression models that may fail to capture nonlinear behavioral relationships, Gradient Boosting

continuously improves prediction performance by correcting previous errors during training. This makes it highly effective for analyzing complex digital behavior patterns where multiple emotional, behavioral, and lifestyle factors interact simultaneously. Its robustness, adaptability, and predictive precision make it ideal for digital wellbeing prediction.

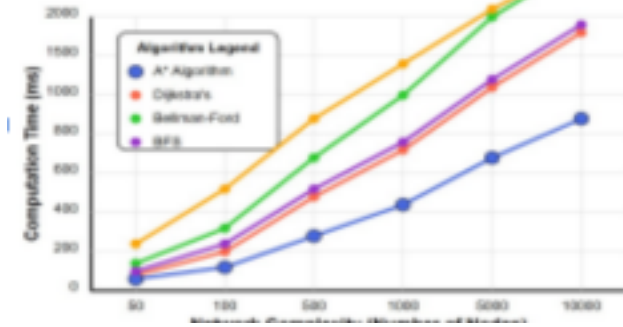


Fig. 2. Algorithm Comparison of ScreenSense

F. What are we implementing?

We are implementing an intelligent Digital Wellbeing Prediction and Behavioral Analytics Platform that helps users understand how their digital habits influence focus, productivity, emotional stability, and overall wellbeing. The system collects behavioral data, performs machine learning-based analysis, generates personalized predictions, and provides actionable recommendations through an interactive dashboard.

G. Why are we implementing?

This platform is developed to address the growing impact of excessive digital engagement, screen addiction, notification overload, and irregular digital routines on mental health and productivity. Many users are unaware of how their daily digital habits influence concentration, sleep quality, and emotional balance. ScreenSense aims to create awareness,

promote healthier digital practices, and support users in maintaining balanced technology usage through data-driven insights and predictive analytics.

H. How are we implementing?

The system is built using a modular architecture to ensure scalability, flexibility, and easy maintenance of different components. Key features include behavioral data tracking, machine learning-based digital

wellbeing prediction, and interactive analytics dashboards for personalized insights. The platform also integrates secure user authentication, real-time data processing, and AI-driven recommendations to help users understand and improve their digital habits effectively.

III. RELATED WORKS

The following studies are relevant to our ScreenSense platform:

1) Digital Behaviour and Smartphone Usage Patterns in Wellbeing Prediction Systems [1]

This study analyzes how smartphone usage patterns influence user wellbeing and productivity. It focuses on behavioral indicators such as screen time, app switching frequency, notification overload, and sleep disruption. The research highlights that excessive digital engagement is strongly linked to reduced concentration and increased mental fatigue, especially among young users. The study uses data-driven analysis techniques to identify behavioral trends and correlates them with emotional and cognitive performance. It concludes that continuous monitoring of digital activity can help in understanding user behavior more effectively and support the development of predictive wellbeing systems similar to ScreenSense.

2) Machine Learning Approaches for Mental Health and Behaviour Prediction [2]

This paper presents a machine learning-based framework for analyzing behavioral and emotional data to predict mental health conditions. The study explores different supervised learning algorithms such as Random Forest, Support Vector Machines, and Regression models for prediction tasks. The research emphasizes the importance of feature engineering, as behavioral attributes like activity level, usage patterns, and lifestyle indicators significantly impact model accuracy. The study concludes that machine learning can effectively identify hidden patterns in behavioral data and provide meaningful predictions, which aligns closely with the prediction approach used in ScreenSense.

3) Behavioral Analytics in Digital Wellbeing Applications [3]

This research focuses on the role of behavioral analytics in improving digital wellbeing. It studies how user interaction data from mobile applications can be analyzed to generate insights about attention span, productivity, and emotional state. The paper introduces

visualization techniques and clustering methods . users based on similar behavioral patterns. It concludes that combining analytics with visualization improves user

4) AI-Based Personalized Recommendation Systems for User Productivity [4]

This study proposes an AI-driven recommendation system that suggests personalized actions based on user behavior. It uses machine learning models to analyze user activity data and generate adaptive recommendations for improving productivity and reducing digital distraction. The system continuously learns from user behavior and updates its recommendations accordingly. The study highlights that adaptive systems significantly improve user engagement and long-term behavioral improvement, which is directly relevant to the recommendation module in ScreenSense.

5) Data-Driven Behaviour Monitoring Systems Using Web and Mobile Platforms [5]

This paper presents a system that collects and analyzes user activity data from web and mobile platforms to understand digital behavior patterns. It integrates data visualization tools and predictive analytics to provide real-time insights into user engagement levels. The study uses a modular system architecture and emphasizes secure data handling and scalable design. It concludes that combining behavioral tracking with analytics improves decision-making and supports the development of intelligent monitoring systems like ScreenSense.

navigation and a consistent user experience across the application.



Fig. 3. Dashboard Page

B. Behavioral Data Input and Tracking Flow

The Behavioral Data Input module allows users to provide key digital activity details required for analysis. This includes inputs such as screen time, application usage patterns, notification frequency, sleep duration, focus level, mood score, and anxiety level. All inputs are validated to ensure correctness and consistency before processing. The collected data is then stored securely and used for further analysis in the machine learning pipeline. This module forms the foundation of the ScreenSense system by enabling accurate behavioral understanding of the user.

C. Machine Learning Prediction Flow

The prediction module is responsible for analyzing user behavior and generating a digital wellbeing score. Once the behavioral data is collected, it undergoes preprocessing steps such as cleaning, normalization, and feature selection. Multiple machine learning models including Linear Regression, Random Forest, Support Vector Regression, and Gradient Boosting are trained and evaluated. The system finally selects the best-performing model (Gradient Boosting) for real-time prediction.[2] The output includes a predicted wellbeing score along with performance indicators that help users understand,

IV. RESULTS AND DISCUSSIONS

A. Login and Signup Page Flow

ScreenSense is designed as a secure and user-friendly platform that begins with an authentication system consisting of login and signup functionalities. Users are required to create an account or log in using their credentials, which are securely stored and managed in the backend database. This ensures data privacy and personalized access to the platform. After successful authentication, users are directed to the main dashboard, where they can access different features of the system such as behavioral tracking, digital wellbeing prediction, analytics. This structured flow ensures smooth

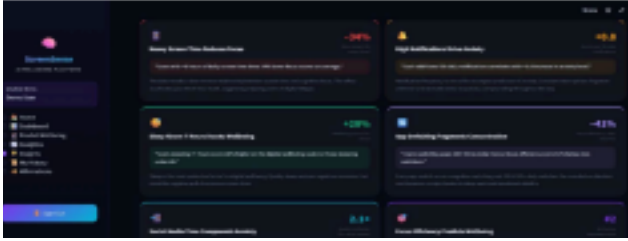


Fig. 4. Prediction wellbeing Page Flow



Fig. 5. Prediction Page

D. Analytics, Visualization, and Recommendation Flow

The analytics module provides users with interactive dashboards that display behavioral trends and patterns. Visualizations such as graphs, correlation plots, and clustering results help users understand how their digital habits affect productivity and mental wellbeing. Additionally, the system generates AI-based personalized recommendations based on user behavior. These suggestions help users reduce screen time, improve focus, and maintain a balanced digital lifestyle.[4]The system also supports PDF report generation, allowing users to track their progress over time.

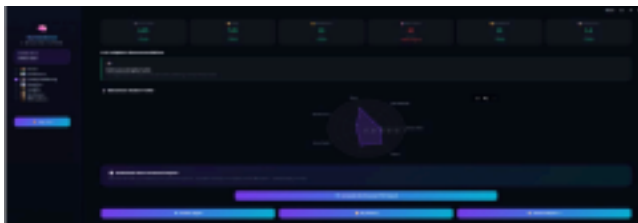


Fig. 6. Insight and report generation page

E. Key Components:

1) Streamlit Framework

The frontend of ScreenSense is developed using Streamlit, which provides an interactive and

The backend is implemented using Python, which handles data processing, machine learning model execution, authentication logic, and report generation.

responsive web interface for displaying dashboards,

2) Python Backend

The system uses multiple supervised learning algorithms such as Linear Regression, Random Forest, Support Vector Regression, and Gradient Boosting to analyze behavioral data and predict digital wellbeing scores.

3) Data Processing Libraries

Libraries like Pandas and NumPy are used for data preprocessing, feature engineering, and numerical computation to ensure accurate model training.

4) Visualization Tools

Plotly and Matplotlib are used to generate interactive charts and visual insights that help users understand behavioral trends effectively.

F. Functional Flow:

1) User Input

The user logs into the system and enters behavioral data such as screen time, sleep duration, notification count, and mood-related parameters

2) Data Processing

The input data is validated and preprocessed to remove inconsistencies and prepare it for model prediction.

3) Model Prediction

The processed data is passed through the trained machine learning model, which predicts the digital wellbeing score based on learned patterns.

4) Result Generation

The system generates the final prediction along with insights, behavioral analysis, and personalized recommendations.

V. CONCLUSION

Currently, there is a growing concern regarding excessive digital usage and its impact on users' mental health, productivity, and emotional stability. However, most existing applications do not provide a unified system that can track digital behavior, analyze it using machine learning, and generate personalized wellbeing insights in a single platform. To address this gap, ScreenSense has been developed as an intelligent digital wellbeing prediction system that combines behavioral analytics and machine learning techniques. The platform is designed to help users understand how their daily digital habits affect their focus, productivity, sleep patterns, and overall wellbeing. By providing real-time predictions and personalized insights, ScreenSense encourages users to develop healthier digital routines. In its current version, the system focuses on behavioral tracking, wellbeing prediction, and interactive analytics dashboards. In the future, ScreenSense can be extended with additional features such as deeper emotional analysis, advanced habit tracking, and integration with wearable devices. These enhancements will further improve accuracy and provide a more complete understanding of user wellbeing. Overall, ScreenSense aims to deliver a simple, effective, and scalable solution for managing digital wellbeing through data-driven insights and intelligent prediction systems.

VI. FUTURE SCOPE

The current version of ScreenSense provides core functionalities such as behavioral tracking, machine learning-based wellbeing prediction, and personalized analytics. However, several improvements and advanced features can be introduced in future versions to enhance its capabilities. One major enhancement will be the inclusion of real-time behavioral tracking, where user activity can be continuously monitored to provide instant feedback on digital habits. Another important feature is emotion recognition integration, which can analyze user mood more accurately using advanced data signals. In addition, the platform can be extended with a mobile application version, allowing users to track their digital wellbeing on smartphones more conveniently. Integration with wearable devices such as smartwatches can also be introduced to collect physiological data like sleep quality, heart

rate, and activity levels for more accurate predictions. A personalized AI recommendation system can further be improved to provide adaptive suggestions based on long-term behavioral patterns. The system can also include a gamification feature, encouraging users to reduce screen time through goals, rewards, and progress tracking.

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